25900695 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) Regular EXAMINATION October 2025

Third SEMESTER

Discipline Specific Elective (DSE) - MG3DSEZIM201 - BIO ANALYTICAL TECHNIQUES IN MICROBIOLOGY AND IMMUNOLOGY

(2024 ADMISSION ONWARDS)

Duration: 2 Hours **Maximum Marks: 70**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Fill in the blanks

Answer all questions

Each question carries **1** marks

1. The device inside a centrifuge that holds tubes is called a ..... [K] / [CO1]
2. Thin Layer Chromatography is a ..... type. [U] / [CO4]
3. Northern blotting is a technique used for the detection of specific .....molecule. [K] / [CO3]
4. The part of the antigen recognized by the antobody is called..... [K] / [CO6]
5. Laminar air flow cabinets are used in .....laboratories. [K] / [CO1]
6. The acid fast staining is used to detect.....species. [K] / [CO2]
7. In .....PAGE ,Sodium dodecyl sulfate denatures proteins and gives them a uniform negative charge. [U] / [CO4]
8. In western blotting ,proteins are transferred onto a.....memberane. [U] / [CO3]

9. Widal test detect antibodies aganist bacteria..... [U] / [CO6]

10. An.....is any substance that can induce an immune response. [U] / [CO6]

[1x10 = 10]

Part B

Short Answer Questions

Answer any **14** questions

Each question carries **2** marks

11. Compare the reliability of pH meter readings in comparison to pH paper. [K] / [CO1]

12. Explain why is pencil used in chromatography instead of a pen to draw base line. [U] / [CO4]

13. Compare RAPD with RFLP in terms of speed and cost. [U] / [CO3]

14. Name two types of agglutination reaction. [K] / [CO6]

15. Differentiate between simple microscope and compound microscope. [U] / [CO2]

16. Explain why basic stains are commonly used for bacterial staining. [U] / [CO2]

17. State the principle of a spectrophotometer. [K] / [CO4]

18. Differentiate between denaturation and extension step in PCR. [K] / [CO3]

19. Define passive immunity. Give one example for naturally acquired passive immunity. [U] / [CO6]

20. Explain the principle of ELISA. [U] / [CO6]

21. Define inoculation. [K] / [CO1]

22. Define Retention time in chromatography. [U] / [CO4]

23. Justify the use of AFLP over SSR markers in biodiversity studies. [U] / [CO3]

24. Write short note on polyvalent vaccine. [U] / [CO6]

25. Name two surface markers present on NK cells. [K] / [CO5]

26. Mention any two disadvantage of a phase contrast microscopy. [K] / [CO2]

[2x14 = 28]

Part C

Short Essay Questions

Answer any **8** questions

Each question carries **4** marks

27. Outline the historical development of planar chromatography ,how has it involved and impacted on modern chromatography.	[A] / [CO4]
28. Explain the application of RFLP in detecting genetic disorders,paternity testing forensic science and other lant breeeding programs.	[A] / [CO3]
29. Differentiate between primary and secondary lymphoid organs .	[U] / [CO5]
30. Describe the principle and procedure of capsule staining.	[U] / [CO2]
31. Differentiate between temperature regulation in an autoclave and a hot air oven.	[K] / [CO1]
32. Compare column chromatography with thin layer chromatography.	[U] / [CO4]
33. Explain the application of DNA finger printing in various biological fields.	[A] / [CO3]
34. Differentiate between CD4+ and CD 8+ T cells.	[K] / [CO5]
35. Write an essay on cells of the immune system.	[K] / [CO5]
36. Illustrate the working principle of a SEM with a neat labelled diagram.	[U] / [CO2]

[4x8 = 32]